

EKAM GHOTRA

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Experience

Software Engineer Intern, Intelligence

May 2025 - Current

Enterprise Mobility - Integral

Madison, WI

- Migrated REST APIs from **Jersey/Guice** to **Spring Boot 3**, modernizing service layers to improve build times and streamline dependency injection
- Developed an Automated VIN Capture Tool with **Tesseract-OCR** and a **TensorFlow** workflow that flags low-confidence reads with a **Bayesian** threshold, aiding a 70% reduction in manual data entry
- Stored VIN-image hashes, capture time, and confidence scores in a **MongoDB** collection, providing adjusters ease of search for prior damage photos and reducing lookup time

Machine Learning Engineer Intern

Sep 2024 - Dec 2024

Equii Foods

Remote

- Part-Time Intern, fine-tuning **Random Forest** classifiers in **scikit-learn** to detect outlier proteins in prototype recipes, achieving **0.92 AUC** and a false positives reduction

Data Engineer Intern

Jun 2024 – Aug 2024

Comcast - Engineering Operations

Livermore, CA

- Analyzed field-active tech telemetry to pinpoint efficiency bottlenecks and reliability risks, collaborating with analysts and engineers to inform targeted process improvements
- Engineered **ELT** pipelines with **Apache Airflow**, extracting from **Microsoft** and **Oracle SQL** servers into **Snowflake** and automating **Tableau** extracts, contributing to a **20%** improvement in data processing and reporting time
- Developed a data-quality framework using **scikit-learn**'s **Random Forest** classifier to detect anomalies and missing values, integrated with **Airflow** to send automated alerts, contributing to on-team manual triage reduction of **30%**

Projects

Portfolio Analytics Toolkit | *Python - Pandas, NumPy, SciPy*

May 2025 - Current

- Leveraged **Pandas** and **NumPy** to implement **Time Series** analysis, calculating basic & annualized returns, Sharpe ratio, drawdowns, skewness, and kurtosis
- Utilized **SciPy** for **Monte Carlo** simulations to estimate VaR, Conditional VaR, and portfolio value projections
- Built a **Markowitz** optimization module to compute asset covariance, portfolio variance, and Sharpe ratios, using **SciPy** to identify the efficient frontier's optimal two-asset mix

Social Media Sentiment Trader | *Python - Pandas, NumPy, finBERT, SQLite*

Jan 2025 - Current

- Pulled tweets daily, scoring them with a finance-tuned sentiment model, leveraging **FinBERT** to flag the strongest bullish/bearish chatter
- Translated high-confidence signals into automated long/short order on referenced ticker, routing the trade to a paper-trading sandbox
- Logged trades to **SQLite** and displayed running returns and drawdown in a **Jupyter Notebook** dashboard

Technical Skills

Languages: Python, SQL, C/C++, Java, R

Frameworks/Libraries: Pandas, NumPy, SciPy, Stasmodels, PyPortfolioOpt, TensorFlow, PyTorch, TA-Lib, Scikit-Learn, Keras

Tools: MATLAB, VSCode, IntelliJ, RStudio, Airflow, Snowflake, Azure, AWS, BigQuery, MongoDB, Tableau, Docker

Education

University of Wisconsin - Madison

Sep 2022 – May 2026

BS Computer Science + BS Data Science, Minor: Economic Analytics

Madison, WI

- **Concentration:** Software Development, Model Making, Statistical Techniques
- **Relevant Course Topics:** Data Structures & Algorithms, Object Oriented Programming, Econometrics, Data Analytics for Economists, Statistical Modeling, Linear Algebra, Software Engineering, Machine Learning & Neural Networks, Data Pipelines, Discrete Mathematics
- **Student Organizations:** Founder - CodeForGood, Cardinal Trading Group, LeetCode Club, Data Science Club

Undecided

Aug 2026 - Jun 2027

Masters of Science

- **Undecided Concentration:** Financial Engineering, Financial Mathematics, or Applied Mathematics & Computation